A WEIGHTED TURÁN SIEVE FOR TWIN PRIMES VIA THE KRAFFT GEOMETRY

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Abstract. The Twin Prime Conjecture is notoriously obstructed by the parity barrier, which prevents classical multiplicative sieves from isolating prime pairs. In this paper, we introduce an additive Turán sieve framework evaluated over bounded, symmetric intervals. Utilizing a centered modular alignment (which we term the Krafft alignment), we construct an additive sieve formulation where the existence of twin primes corresponds to the vanishing of a local penalty function. By evaluating the character sums associated with this sieve, we isolate a destructive interference term at the evaluation point $h = 3q/p$. We subsequently reduce the Twin Prime Conjecture to a variational optimization problem over a finite-dimensional parameter space: the existence of twin primes in a prescribed interval is guaranteed, provided the minimum sieve weight quotient satisfies $\mu_{\min}(n) < 1$. We further show that independent, one-dimensional sieve weights are strictly insufficient to overcome the logarithmic divergence of Mertens' sums, forcing the quotient to $\mu \geq 1$. This structural deficit constitutes a spectral manifestation of the Selberg parity barrier, demonstrating that any successful sieve must intrinsically rely on higher-order correlations and multidimensional exponential sums. All discrete definitions and structural lemmas in this paper have been formally verified in Lean 4.

1. Introduction

The twin prime conjecture is notoriously obstructed by the parity barrier, which prevents classical sieve methods from isolating prime pairs. In this paper, we develop a new additive Turán sieve framework [17] designed to bypass this combinatorial obstruction. Rather than relying on traditional multiplicative inclusion-exclusion, we map the search for twin primes to an additive penalty function over a highly structured, symmetric interval—utilizing a centered modular alignment (which we term the Krafft alignment) [9, 13]. By analyzing the spectral properties of this penalty function, we reduce the twin prime conjecture to a finite-dimensional variational optimization problem.

To understand the necessity of this approach, we must consider the historical context of the parity barrier. Classical benchmarks for prime density were established by

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the Hardy-Littlewood conjecture [8]. Subsequent sieve methods, originating with Brun [2] and refined by Selberg [14] and others [11], have provided the main avenues of attack on prime distribution. While these frameworks have yielded extraordinary structural results—most notably the bounded gaps between primes [3, 4, 6, 10]—they inevitably encounter Selberg's parity barrier. This phenomenon demonstrates that classical sieves cannot fundamentally distinguish between integers possessing an even number of prime factors and those possessing an odd number, precluding a direct proof of the twin prime conjecture using these methods alone.

By evaluating the character sums associated with this discrete sieve over the cyclic group via the discrete Fourier transform and Plancherel's theorem, we show that the Krafft alignment inherently forces destructive interference at the evaluation point corresponding to $h = 3q/p_i$. Using a compactness argument over the space of multidimensional sieve weights, we define a minimum sieve weight quotient, $\mu_{\min}(n)$, reducing the twin prime problem to a finite-dimensional optimization: if the bound $\mu_{\min}(n) < 1$ holds, twin prime indices must exist within $\mathcal{A}_n$.

Furthermore, we propose a variational optimization of the sieve weights to minimize the penalty contribution. By treating the weights as a convex optimization problem in the frequency domain, we show that independent, one-dimensional sieve weights are structurally insufficient to overcome the local density bounds. Because isolated character sum evaluations cannot outpace the logarithmic divergence of Mertens' sums, the sieve quotient is strictly forced to $\mu \geq 1$. This analytic deficit reveals that resolving the conjecture requires exploiting the full multidimensional space of higher-order correlations.

The paper is organized as follows. In Section 2, we formalize the Krafft geometry and define the additive sieve. Section 3 shifts the problem into the frequency domain, evaluating the necessary discrete Fourier transforms and identifying the structural destructive interference. Section 4 proves the spectral manifestation of the parity barrier, showing that one-dimensional weights are insufficient. Section 5 establishes the continuous variational framework and proves that the minimization of our sieve quotient $\mu_{\min}(n)$ guarantees the existence of twin primes. Finally, Section 6 explores the full character convolution and presents numerical optimizations that empirically validate the structural necessity of multidimensional correlations.

Throughout the text, symbols are used as direct links to formalized code in the GitHub repository `github.com/ElNando888/KrafftSieve/tree/v1.4`, which contains the machine-checked verifications for all definitions, statements, and proofs in this paper.[1] We use Lean [5] version 4.28.0 together with the Lean mathematical library `Mathlib` [16] revision `8f9d9cf` (dated 2026-02-16).

[1]The only axioms used are `propext`, `Classical.choice`, and `Quot.sound`, which can be checked by the `#print axioms` command.

2. Discrete geometry and the additive sieve equivalence

To circumvent the parity limitations inherent in multiplicative sieves, we transition to an additive framework evaluated over a bounded, symmetric interval. We begin by formalizing the specific geometry of our prime window and sieve domain.

Definition 2.1 (The Prime Window and Primorial [&])**.** *Let $\mathcal{P}_n$ denote the set of primes evaluated by the sieve, bounded as:*

$$\mathcal{P}_n = \{p \text{ prime} \mid 5 \leq p < 6n + 2\}$$

We define $w(n) = |\mathcal{P}_n|$ as the total count of these prime factors, and $q = \prod_{p \in \mathcal{P}_n} p$ as their associated primorial.

The underlying geometric transformation we employ is rooted in an arithmetic equivalence first noted by the astronomer W. L. Krafft in 1798 [9]. While the Diophantine form $6ab \pm a \pm b$ itself remains relatively obscure, the equivalence between twin prime pairs and the sifting of natural indices avoiding these forms was notably acknowledged by W. Sierpiński in his selection of number theory problems [15] (and is documented as OEIS sequence A002822 [12]). We formalize this historically noted mapping as a discrete residue condition.

Definition 2.2 (The Krafft Residues [ ])**.** *For each prime $p_i \in \mathcal{P}_n$, we establish the target Krafft alignment residue as:*

$$r_i^K = \left\lfloor \frac{p_i + 1}{6} \right\rfloor$$

Definition 2.3 (The Evaluation Interval [ ])**.** *We constrain our search for twin prime indices to the highly structured symmetric interval $\mathcal{A}_n$:*

$$\mathcal{A}_n = [6n^2 - 2n, 6n^2 + 10n + 3]$$

This specific interval width is chosen so that the alignment residue r_i^K effectively translates the standard divisibility conditions of twin prime candidates into a single modular congruence evaluated at the index x. The following lemma makes this equivalence precise.

Lemma 2.4 (Modular Alignment Equivalence [ ])**.** *For any prime factor $p \geq 5$ and any integer x, let $r = \lfloor (p+1)/6 \rfloor$. Then $x \equiv \pm r \pmod{p}$ if and only if $p \mid (6x - 1)$ or $p \mid (6x + 1)$.*

Proof. Since $p \geq 5$ is prime, we have either $p \equiv 1 \pmod 6$ or $p \equiv 5 \pmod 6$. If $p \equiv 1 \pmod 6$, then $r = (p-1)/6$, and therefore $6r = p - 1 \equiv -1 \pmod{p}$. If $p \equiv 5 \pmod 6$, then $r = (p+1)/6$, yielding $6r = p + 1 \equiv 1 \pmod{p}$. In both cases, $6r \equiv \pm 1 \pmod{p}$. Assuming $x \equiv \pm r \pmod{p}$, multiplying both sides by 6 gives $6x \equiv \pm 6r \equiv \pm 1 \pmod{p}$. This implies $6x \mp 1 \equiv 0 \pmod{p}$, meaning p divides either $6x - 1$ or $6x + 1$. The steps are reversible, establishing the equivalence. $\square$

To track the sieve's progress additively rather than through a multiplicative inclusion-exclusion product, we establish an indicator function for the forbidden residue classes and a global penalty (or counting) function that accumulates across all primes in our window.

Definition 2.5 (Local Indicator and Global Penalty Function [&])**.** *We define the local indicator function $g_i(x) \in \{0, 1\}$ to detect whether an integer x falls into the forbidden residue classes modulo p_i:*

$$g_i(x) = \begin{cases} 1, & \textit{if } x \equiv \pm r_i^K \pmod{p_i}; \\ 0, & \textit{otherwise.} \end{cases}$$

The global penalty function $c(x)$ is defined additively as the sum of all local indicators across the prime window:

$$c(x) = \sum_{i=1}^{w(n)} g_i(x)$$

Unlike classical inclusion-exclusion setups which track the surviving elements via alternating signs, $c(x)$ is non-negative and simply counts the number of forbidden congruences satisfied by x. An integer with zero penalty ($c(x) = 0$) has successfully avoided all prime factors in $\mathcal{P}_n$. To ensure that avoiding these specific prime factors is sufficient to guarantee true primality, we rely on the strict algebraic boundaries of the interval $\mathcal{A}_n$.

Lemma 2.6 (Interval Projection Bound [ ])**.** *For any integer $x \in \mathcal{A}_n$, the value $6x + 1$ is strictly less than the square of the next possible prime candidate after the sieve window, namely $6x + 1 < (6n + 5)^2$.*

Proof. The maximum value in the interval $\mathcal{A}_n$ is $x = 6n^2 + 10n + 3$. Evaluating $6x + 1$ at this maximum yields $6(6n^2 + 10n + 3) + 1 = 36n^2 + 60n + 19$. The next prime candidate strictly greater than the upper bound of $\mathcal{P}_n$ (which is $6n + 1$) is at least $6n + 5$. Comparing our maximal projection to the square of this candidate gives $(6n + 5)^2 = 36n^2 + 60n + 25$. Since $36n^2 + 60n + 19 < 36n^2 + 60n + 25$, the strict inequality $6x + 1 < (6n + 5)^2$ holds for all $x \in \mathcal{A}_n$. $\square$

This bound is key to the reduction, as it ensures that any survivors within the interval are forced to be prime.

Theorem 2.7 (Additive Sieve Formulation [ ])**.** *For any integer $n \geq 1$ and any $x \in \mathcal{A}_n$, the global penalty function evaluates to exactly zero, $c(x) = 0$, if and only if both $6x - 1$ and $6x + 1$ are prime numbers.*

Proof. By the definition of the global penalty function, $c(x) = 0$ if and only if $g_i(x) = 0$ for all $1 \leq i \leq w(n)$. This means $x \not\equiv \pm r_i^K \pmod{p_i}$ for all primes $p_i \in \mathcal{P}_n$. By Lemma 2.4, this guarantees that neither $6x - 1$ nor $6x + 1$ has any prime divisors in $\mathcal{P}_n$. Integers of the form $6x \pm 1$ are intrinsically coprime to 2 and 3. Therefore,

$6x-1$ and $6x+1$ have no prime factors strictly less than $6n+5$. Because Lemma 2.6 structurally guarantees that the maximum possible value of $6x+1$ across the entire interval $\mathcal{A}_n$ is strictly less than $(6n+5)^2$, any composite number of the form $6x \pm 1$ must have a prime factor strictly less than $6n+5$. Since all such factors have been eliminated, it follows that both $6x-1$ and $6x+1$ must be prime. □

3. Character sum expansion and the main sieve identity

To analyze the distribution of twin prime indices $x \in \mathcal{A}_n$ that avoid the local penalties, we first construct a weighted sifting condition in the spatial domain, which we will subsequently evaluate through its character sums over the cyclic group. We begin by introducing an arbitrary, real-valued weight function $W(x)$ strictly supported within our bounded evaluation interval $\mathcal{A}_n$.

Definition 3.1 (Sieve Weight Sums)**.** *For an arbitrary real-valued weight function $W(x)$ supported strictly on $\mathcal{A}_n$, we define the total weight sum S_1 and the penalized weight sum S_2:*

$$S_1(n, W) = \sum_{x \in \mathcal{A}_n} W(x)$$

$$S_2(n, W) = \sum_{x \in \mathcal{A}_n} W(x)c(x)$$

Lemma 3.2 (The Weighted Existence Principle [[illegible]])**.** *If there exists a non-negative weight function $W(x)$ such that the global penalized sum is strictly less than the base weight sum, $S_2(n, W) < S_1(n, W)$, then there must exist at least one integer $x \in \mathcal{A}_n$ such that $W(x) > 0$ and $c(x) = 0$.*

Proof. The global penalty function $c(x)$ is a sum of non-negative indicator functions, forcing $c(x) \geq 0$. Consequently, any non-zero penalty must satisfy $c(x) \geq 1$. If $c(x) \geq 1$ for all coordinates where $W(x) > 0$, then the sum $S_2(n, W) = \sum W(x)c(x)$ must be greater than or equal to $S_1(n, W) = \sum W(x)$. This contradicts the strict inequality $S_2 < S_1$. Therefore, there must exist at least one index x satisfying $c(x) = 0$ and carrying strictly positive weight. □

Definition 3.3 (Sieve Sufficiency Condition)**.** *We say the sieve sufficiency condition holds for a given n if there exists a weight function $W : \mathbb{Z}/q\mathbb{Z} \to \mathbb{R}$ that is non-negative everywhere, supported on $\mathcal{A}_n$, and satisfies the strict inequality $S_2(n, W) < S_1(n, W)$.*

Theorem 3.4 (Sieve Sufficiency Guarantee [[illegible]])**.** *For any integer $n \geq 1$, if Definition 3.3 is satisfied, then there exists an integer $x \in \mathcal{A}_n$ such that $6x-1$ and $6x+1$ form a twin prime pair.*

Proof. If Definition 3.3 is satisfied, then Theorem 3.2 guarantees the presence of at least one index $x \in \mathcal{A}_n$ satisfying zero penalty ($c(x) = 0$). Under Theorem 2.7, the vanishing of the penalty implies that both $6x-1$ and $6x+1$ are prime. □

In the spatial domain, analyzing the pseudo-random overlaps of the forbidden residue classes across all primes is analytically intractable. We therefore shift into the frequency domain, where the strict periodicities of the local indicator functions resolve into highly sparse, orthogonal spectra, allowing us to separate the principal density from the oscillatory interference. This is accomplished by expanding our functions over the characters of the cyclic group $\mathbb{Z}/q\mathbb{Z}$.

Definition 3.5 (Fourier transform over $\mathbb{Z}/q\mathbb{Z}$)**.** *For any function $f(x)$, the frequency coefficients of its Fourier transform over $\mathbb{Z}/q\mathbb{Z}$ are given by:*

$$\hat{f}(h) = \frac{1}{q}\sum_{x=0}^{q-1} f(x)e^{-2\pi i hx/q}$$

By exploiting the bounds of $\mathcal{A}_n$, we can extend our spatial sums over the entire cyclic group without altering their values. This allows us to apply Plancherel's theorem to expand the penalized mass into a series of exponential sums.

Lemma 3.6 (Compact Support Equivalence [GR])**.** *Assume the weight function $W(x)$ is supported within the interval $\mathcal{A}_n$, meaning $W(x) = 0$ for all $x \notin \mathcal{A}_n$. Then the total weight $S_1(n, W)$ is proportional to the zero-frequency Fourier coefficient of the weight function:*

$$S_1(n, W) = q\hat{W}(0)$$

Proof. Since $W(x)$ is zero outside $\mathcal{A}_n$, the sum over the interval is equivalent to the sum over the entire cyclic group $\mathbb{Z}/q\mathbb{Z}$. Evaluating Definition 3.5 at $h = 0$ yields

$$\hat{W}(0) = \frac{1}{q}\sum_{x\in\mathbb{Z}/q\mathbb{Z}} W(x)e^0 = \frac{1}{q}S_1(n, W)$$

Multiplying by q completes the proof. □

Lemma 3.7 (The Character Sum Penalty Expansion [GR])**.** *For a weight function $W(x)$ supported on $\mathcal{A}_n$, the weighted penalty sum $S_2(n, W)$ decomposes into the Fourier domain as:*

$$S_2(n, W) = q\sum_{i=1}^{w(n)}\sum_{h=0}^{q-1} \hat{W}(h)\overline{\hat{g}_i(h)}$$

Proof. Substitute the additive definition $c(x) = \sum g_i(x)$ into $S_2(n, W)$. Because $W(x)$ has compact support on $\mathcal{A}_n$, the sum extends to $\mathbb{Z}/q\mathbb{Z}$ without altering the value. Applying the standard discrete Plancherel theorem to $W(x)$ and $c(x)$, and utilizing the linearity of the Fourier transform over the finite sum of local indicators, yields the expansion. □

Because the local indicator function $g_i(x)$ depends only on the residue of x modulo the local prime p_i, projecting it onto the full cyclic group $\mathbb{Z}/q\mathbb{Z}$ yields a highly sparse spectrum.

Lemma 3.8 (Indicator function spectrum [&])**.** *The discrete Fourier transform (as defined in Definition 3.5) of the local indicator function, $\hat{g}_i(h)$, evaluates to zero for all frequencies except those that are precise integer multiples of q/p_i. Specifically, let $h \in \mathbb{Z}/q\mathbb{Z}$:*

(1) *If $h = k(q/p_i)$ for some integer $0 \le k < p_i$, then*

$$\hat{g}_i\left(k\frac{q}{p_i}\right) = \frac{2}{p_i}\cos\left(\frac{2\pi k r_i^K}{p_i}\right)$$

(2) *If h is not a multiple of q/p_i, then $\hat{g}_i(h) = 0$.*

Proof. By writing $x = y + mp_i$ with $0 \le y < p_i$ and $0 \le m < q/p_i$, we can factor the Fourier transform as

$$\begin{aligned}\hat{g}_i(h) &= \frac{1}{q}\sum_{x=0}^{q-1} g_i(x)e^{-2\pi i h x/q} \\ &= \frac{1}{q}\sum_{y=0}^{p_i-1} g_i(y)e^{-2\pi i h y/q}\sum_{m=0}^{q/p_i-1} e^{-2\pi i h m p_i/q}.\end{aligned}$$

The inner sum over m vanishes identically unless h is a multiple of q/p_i. When $h = k(q/p_i)$, the sum over m evaluates to q/p_i, yielding

$$\hat{g}_i\left(k\frac{q}{p_i}\right) = \frac{1}{p_i}\sum_{y=0}^{p_i-1} g_i(y)e^{-2\pi i k y/p_i}.$$

Since $g_i(y) = 1$ only at the two residues $y \equiv \pm r_i^K \pmod{p_i}$, we obtain

$$\begin{aligned}\hat{g}_i\left(k\frac{q}{p_i}\right) &= \frac{1}{p_i}\left(e^{-2\pi i k r_i^K/p_i} + e^{2\pi i k r_i^K/p_i}\right) \\ &= \frac{2}{p_i}\cos\left(\frac{2\pi k r_i^K}{p_i}\right).\end{aligned}$$

□

This sparsity allows us to explicitly separate the principal expected density term from the targeted oscillatory interference, yielding the main identity of the sieve. When evaluating the precise exponential sum at $h = 3q/p_i$, the trigonometric phase aligns with the alignment residue, causing destructive interference.

Lemma 3.9 (Evaluation of the Character Sum at $h = 3q/p_i$ [ ])**.** *For any prime factor $p_i \ge 5$, substituting the alignment residue $r_i^K = \lfloor (p_i+1)/6 \rfloor$ evaluated precisely at $k = 3$ yields a strictly negative term:*

$$\cos\left(\frac{2\pi \cdot 3 \cdot r_i^K}{p_i}\right) = -\cos\left(\frac{\pi}{p_i}\right)$$

Proof. Resolving the floor function $r_i^K = \lfloor (p_i+1)/6 \rfloor$ gives $r_i^K = \frac{p_i \pm 1}{6}$, uniquely determined by the parity of $p_i \pmod 6$. Evaluating the phase angle generates $\frac{6\pi}{p_i}\left(\frac{p_i \pm 1}{6}\right) = \pi \pm \frac{\pi}{p_i}$. Applying the standard trigonometric identity $\cos(\pi \pm \theta) = -\cos(\theta)$ directly yields the result. □

Theorem 3.10 (The Main Sieve Identity [ቫ]). *For any weight function $W(x)$ supported on $\mathcal{A}_n$, the penalized sum isolates into a principal expected density term and an oscillatory interference sum:*

$$S_2(n, W) = S_1(n, W) \sum_{i=1}^{w(n)} \frac{2}{p_i} + q \sum_{i=1}^{w(n)} \sum_{k=1}^{p_i - 1} \hat{W}\left(k \frac{q}{p_i}\right) \frac{2}{p_i} \cos\left(\frac{2\pi k r_i^K}{p_i}\right) \tag{1}$$

Proof. Starting from Lemma 3.7, we partition the inner sum over frequencies h into the zero frequency $h = 0$ and the non-zero frequencies. Applying Lemma 3.8, all frequencies h that are not multiples of q/p_i evaluate to $\hat{g}_i(h) = 0$ and are discarded. The sum collapses entirely to the frequencies $h = k(q/p_i)$. Finally, applying Lemma 3.6 replaces $q\hat{W}(0)$ with the total weight $S_1(n, W)$, establishing the exact equation. □

This destructive interference can suppress $S_2(n, W)$ below the total weight $S_1(n, W)$.

Lemma 3.11 (Isolation of the $h = 3q/p_i$ Term [ቫ]). *For a weight function $W(x)$ whose spectral support is restricted to the base harmonic $h = 0$ and the precise frequencies $h = 3q/p_i$, the sieve identity* (1) *simplifies exactly to:*

$$S_2(n, W) = S_1(n, W) \sum_{i=1}^{w(n)} \frac{2}{p_i} - q \sum_{i=1}^{w(n)} \hat{W}\left(3 \frac{q}{p_i}\right) \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right)$$

Proof. By the spectral restriction of W, all terms in equation (1) where $k \neq 3$ vanish identically. For the surviving $k = 3$ terms, applying Lemma 3.9 inverts the sign of the interference sum, replacing the general cosine term with $-\cos(\pi/p_i)$. □

4. The Parity Barrier in One Dimension

Definition 4.1 (The Sieve Weight Quotient). *To evaluate the structural efficiency of a given weight function, we define the sieve weight quotient $\mu(n, W)$ as the ratio of the penalized weight sum to the total weight sum:*

$$\mu(n, W) = \frac{S_2(n, W)}{S_1(n, W)} \tag{2}$$

By Theorem 3.2, if we can construct a weight function such that $\mu(n, W) < 1$, the sieve sufficiency condition (Definition 3.3) is satisfied.

Having isolated the destructive interference at $h = 3q/p_i$, a natural first attempt is to construct weight functions composed entirely of independent, one-dimensional exponential sums. We can assess the limitations of this approach by examining the isolated spectral capacity of the sieve.

Definition 4.2 (Principal Density and Character Suppression Capacity [&])**.** *For a one-dimensional character weight system evaluated over $\mathcal{P}_n$, the principal expected density is governed by the harmonic sum over the prime window:*

$$\textit{Principal Term}(n) = \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i}$$

The corresponding maximum theoretical suppression available from the targeted frequencies is defined as the suppression capacity:

$$\textit{Suppression Capacity}(n) = \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right)$$

If the suppression capacity could exceed or match the principal density term, a one-dimensional sieve could theoretically suppress the penalized sum S_2 below S_1. However, the geometry of the character sum strictly forbids this.

Lemma 4.3 (The Spectral Deficit [ ])**.** *For any valid prime window $\mathcal{P}_n$, the suppression capacity of any purely one-dimensional character weight system is strictly less than the sieve's principal density term:*

$$\sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right) < \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \tag{3}$$

Proof. For any prime $p_i \geq 5$, the phase angle satisfies $0 < \pi/p_i < \pi/2$, which strictly bounds the cosine function: $\cos(\pi/p_i) < 1$. Multiplying by the strictly positive amplitude $2/p_i$ yields

$$\frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right) < \frac{2}{p_i}.$$

Summing these strict, term-by-term inequalities across the finite non-empty set $\mathcal{P}_n$ preserves the strict inequality:

$$\sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right) < \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i},$$

which completes the proof. $\square$

While the strict inequality itself is algebraically elementary, its significance lies in the asymptotic behavior of the divergence.

Proposition 4.4 (One-Dimensional Parity Barrier)**.** *For any sufficiently large n, utilizing purely independent, one-dimensional exponential sum weights forces the sieve weight quotient to $\mu(n, W) \geq 1$. It is analytically impossible to resolve the twin prime conjecture using strictly one-dimensional variance bounds.*

Proof. By Mertens' theorems [11], the principal density term $\sum \frac{2}{p_i}$ diverges logarithmically as $n \to \infty$. Conversely, the magnitude of the spectral deficit (3), measured by the difference between the principal term and the suppression capacity, can be bounded using the Taylor expansion $\cos(x) \geq 1 - x^2/2$:

$$\text{Deficit} = \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i}\left(1 - \cos\frac{\pi}{p_i}\right) \leq \sum_{p_i \in \mathcal{P}_n} \frac{\pi^2}{p_i^3}$$

Because the sum $\sum p^{-3}$ converges over the primes, the absolute suppression capacity of the one-dimensional character evaluations grows strictly slower than the logarithmically diverging principal density. $\square$

This convergence-divergence disparity forces the quotient $\mu \geq 1$ for any sufficiently large n when utilizing one-dimensional independent evaluations. This spectral deficit is precisely the analytic manifestation of the well-known Selberg parity barrier within our additive framework. (A finite truncation of purely independent character sums will inevitably force the weight function to take negative values at dense overlapping forbidden regions before it can reduce the overall weight quotient below 1.)

This obstruction shows that to circumvent the density deficit and overcome the parity barrier, the sieve weight construction must incorporate multidimensional exponential sums and higher-order correlations.

5. Variational Optimization of the Sieve Weight Quotient

The analytic barrier established in Section 4 shows that utilizing purely independent one-dimensional exponential sums fundamentally fails to bypass the structural density deficit. To constructively overcome this, we must transition to a correlated, multidimensional weight space. We build this space using higher-order correlations of the fundamental character variables.

Definition 5.1 (Multidimensional Basis and Polynomial [&])**.** *For any subset of prime indices $S \subseteq \{1, \dots, w(n)\}$, we define the multidimensional basis function as the product of the targeted trigonometric evaluations:*

$$B_S(x) = \prod_{i \in S} \cos\left(\frac{6\pi x}{p_i}\right)$$

Let λ be a real-valued coefficient vector indexed by the power set of $\{1, \dots, w(n)\}$. We construct the multidimensional polynomial $P(x)$ as a linear combination of these basis functions:

$$P(x) = \sum_{S} \lambda_S B_S(x)$$

The resulting $2^{w(n)}$-dimensional basis is closed under pointwise multiplication, which allows us to ensure the non-negativity of the sieve weight function.

Definition 5.2 (The Multidimensional Weight Function [?])**.** *We define the multidimensional weight function $W_\lambda(x)$ by squaring the polynomial $P(x)$ and strictly restricting its support to the evaluation interval $\mathcal{A}_n$:*

$$W_\lambda(x) = \begin{cases} (P(x))^2 & \text{if } x \in \mathcal{A}_n \\ 0 & \text{otherwise} \end{cases}$$

By casting the weight space as the square of a polynomial, we automatically satisfy the condition $W_\lambda(x) \geq 0$ for all x, avoiding the need to impose infinitely many linear inequality constraints on λ. This formulation translates our sieve sums into classical quadratic forms.

Lemma 5.3 (Reduction to Quadratic Forms [? & ?])**.** *The total weight sum $S_1(n, W_\lambda)$ and the penalized weight sum $S_2(n, W_\lambda)$ map directly to continuous quadratic forms $Q_1(\lambda)$ and $Q_2(\lambda)$. Specifically, the total weight sum evaluates to the squared norm of the polynomial over the interval:*

$$S_1(n, W_\lambda) = Q_1(\lambda) = \sum_{x \in \mathcal{A}_n} (P(x))^2$$

Proof. By definition, $S_1 = \sum_{x \in \mathcal{A}_n} W_\lambda(x)$. Substituting the construction $W_\lambda(x) = (P(x))^2$, we expand the square and exchange the order of summation:

$$\begin{aligned} S_1(n, W_\lambda) &= \sum_{x \in \mathcal{A}_n} \left(\sum_S \lambda_S B_S(x) \right)^2 \\ &= \sum_{x \in \mathcal{A}_n} \sum_{S,T} \lambda_S B_S(x) B_T(x) \lambda_T \\ &= \sum_{S,T} \lambda_S \left(\sum_{x \in \mathcal{A}_n} B_S(x) B_T(x) \right) \lambda_T. \end{aligned}$$

Defining the matrix elements $M_1(S,T) = \sum_{x \in \mathcal{A}_n} B_S(x) B_T(x)$ yields the quadratic form $Q_1(\lambda) = \sum_{S,T} \lambda_S M_1(S,T) \lambda_T$.

The derivation for $S_2(n, W_\lambda)$ proceeds identically by incorporating the linear penalty function $c(x)$ into the inner sum:

$$\begin{aligned} S_2(n, W_\lambda) &= \sum_{x \in \mathcal{A}_n} c(x) \left(\sum_S \lambda_S B_S(x) \right)^2 \\ &= \sum_{S,T} \lambda_S \left(\sum_{x \in \mathcal{A}_n} c(x) B_S(x) B_T(x) \right) \lambda_T, \end{aligned}$$

yielding the quadratic form $Q_2(\lambda)$. $\square$

With both sums expressed as quadratic forms, the sieve weight quotient (2) transitions to a Rayleigh-Ritz quotient $R(\lambda) = Q_2(\lambda)/Q_1(\lambda)$ []. Because this quotient is inherently scale-invariant, we can apply a standard compactness argument.

Lemma 5.4 (Compactness of the Orthogonal Unit Sphere []). *Let* $\ker(Q_1)$ *denote the kernel of the quadratic form* Q_1*. Because the quotient is invariant under both uniform scaling and the additive translation of vectors from* $\ker(Q_1)$*, we restrict the parameter space strictly to the unit sphere within the orthogonal complement, denoted as* $\mathcal{S}^{\perp}$*. This bounded subspace* $\mathcal{S}^{\perp}$ *is compact.*

Proof. Within the finite-dimensional Euclidean space of the coefficients, compactness is equivalent to the subspace being both closed and bounded (Heine-Borel theorem). The orthogonal complement $\ker(Q_1)^{\perp}$ forms a closed linear subspace. The unit sphere condition, enforced by the standard dot product $\lambda \cdot \lambda = 1$, defines a set that is closed and bounded. Because $\mathcal{S}^{\perp}$ is the intersection of these sets, it is compact. □

Having established Lemma 5.4, we can now rigorously analyze the continuous image of this subspace under the quotient mapping. This allows us to formally define the absolute lower bound of the sieve weight quotient (2).

Definition 5.5 (The Minimum Attainable Ratio). *We define the set of attainable ratios as the continuous image of the compact space* $\mathcal{S}^{\perp}$ *under the mapping* $R(\lambda)$*. The minimum sieve quotient* $\mu_{\min}(n)$ *is defined as the absolute infimum of this attainable set.*

By using the topological compactness derived above, we can assert the existence of an exact optimal coefficient vector.

Theorem 5.6 (Existence of the Exact Minimizer []). *The minimum sieve quotient* $\mu_{\min}(n)$ *is attained by an optimal coefficient vector* $\lambda_{\text{opt}} \in \mathcal{S}^{\perp}$*, such that* $Q_1(\lambda_{\text{opt}}) > 0$ *and* $R(\lambda_{\text{opt}}) = \mu_{\min}(n)$*.*

Proof. The quotient $R(\lambda)$ is continuous wherever $Q_1(\lambda) > 0$. By confining our domain to the orthogonal complement $\mathcal{S}^{\perp}$, we exclude the kernel of Q_1, ensuring continuity across the entire restricted domain. Because continuous functions map compact spaces to compact spaces, the resulting set of attainable ratios is itself compact. By the extreme value theorem, a compact subset of the real numbers contains its infimum, guaranteeing the existence of the minimizer λ_{opt}. □

With the existence of the minimiser established, we obtain the main result of this section. We can now formally link the continuous bound $\mu_{\min}(n)$ back to the discrete existence of twin prime indices within our evaluation interval.

Theorem 5.7 (Optimal Sufficiency and Sieve Guarantee [&]). *The optimal multidimensional weight function* $W_{\text{opt}}(x)$ *constructed from* λ_{opt} *satisfies the sufficiency condition* ($S_2 < S_1$) *if and only if* $\mu_{\min}(n) < 1$*. If this analytic bound holds, the framework rigorously guarantees the existence of an index for a twin prime pair within* $\mathcal{A}_n$*.*

Proof. By the definition of the minimum ratio, evaluating the moments with the optimal weight function yields exactly $S_2(n, W_{\text{opt}})/S_1(n, W_{\text{opt}}) = R(\lambda_{\text{opt}}) = \mu_{\min}(n)$. Since S_1 is strictly positive for the optimal non-zero weight, it follows algebraically that $S_2 < S_1$ holds if and only if the ratio $\mu_{\min}(n) < 1$. If this strict inequality is satisfied, the weight function W_{opt} inherently fulfills the sufficiency condition of Definition 3.3. By directly applying Theorem 2.7, this sufficiency implies the existence of at least one index $x \in \mathcal{A}_n$ that generates a twin prime pair. □

This theorem completes the geometric reduction. The parity obstruction has been translated into an analytic condition: evaluating the spectral density of the higher-order correlations to confirm $\mu_{\min}(n) < 1$.

6. Discussion: higher-order correlations and future directions

While the reduction of the twin prime conjecture to the spectral bound $\mu_{\min}(n) < 1$ provides a rigid finite-dimensional framework, achieving this bound analytically remains obstructed by the spectral deficit. However, the multidimensional parameter space $\mathcal{S}^{\perp}$ theoretically bypasses this limitation. We can map the precise analytic mechanism of this bypass by examining the full inclusion-exclusion convolution in the frequency domain.

6.1. The perfect sieve weight and character convolution.

To understand how higher-order correlations circumvent the parity barrier, consider the "perfect" spatial weight function $\omega(x)$, which equals 1 when $c(x) = 0$ and 0 elsewhere. It is given by the pointwise product of the allowed residue indicators:

$$\omega(x) = \prod_{i=1}^{w(n)} \big(1 - g_i(x)\big)$$

Transforming this identity into the frequency domain over $\mathbb{Z}/q\mathbb{Z}$ yields a spectral characterization. Let $\hat{g}_i(h)$ be the Fourier transform of the individual indicator functions. By the convolution theorem, the sequence of allowed residues transforms as $q\delta_{h,0} - \hat{g}_i(h)$. Consequently, the perfect multidimensional weight function in the frequency domain resolves into the full character convolution:

$$\Omega_h = \big(q\delta_{h,0} - \hat{g}_1(h)\big) \circledast \big(q\delta_{h,0} - \hat{g}_2(h)\big) \circledast \cdots \circledast \big(q\delta_{h,0} - \hat{g}_{w(n)}(h)\big)$$

where $\circledast$ denotes the circular discrete convolution over $\mathbb{Z}/q\mathbb{Z}$.

When this convolution is expanded, it naturally generates higher-order cross-correlations (e.g., $\hat{g}_i(h_1) \circledast \hat{g}_j(h_2)$). These multidimensional exponential sums appear to be indispensable. Where isolated character evaluations fail by allowing negative functional overlaps, the cross-correlations intuitively act as precise corrective phases. We expect them to constructively interfere at the intersections of forbidden congruences, forcing the weight function to vanish. This multidimensional scaling is thus intended to simultaneously preserve non-negativity and protect the principal density term from over-subtraction, an effect which is empirically supported by numerical optimization of the quotient for small values of n.

6.2. **Numerical explorations of higher-order correlations.** To empirically validate the structural necessity of these multidimensional corrective phases, we implemented the continuous variational framework and computationally minimized the sieve quotient $\mu_{\min}(n)$ via the generalized eigenvalue problem derived from the discrete penalty function. The Python scripts developed for these spectral optimizations are publicly available in the accompanying GitHub repository.

Due to the exponential growth of the basis set of multidimensional sieve weights, a direct calculation incorporating all prime factors and unrestricted correlation depths rapidly exceeds standard computational resources. Therefore, our initial investigations restricted the optimization to two-dimensional correlations (evaluating interference strictly between pairs of primes). Under this restriction, we obtained admissible quotients $\mu_{\min}(n) < 1$ for all bounded ranges up to $n = 260$. Figure 1 illustrates the trajectory of the optimized 2D quotient. While the 2D correlations initially provide sufficient destructive interference to suppress the quotient below 1, the distinct upward trend suggests that the spectral deficit induced by the one-dimensional Mertens terms ultimately overcomes strictly pairwise interactions as n grows.

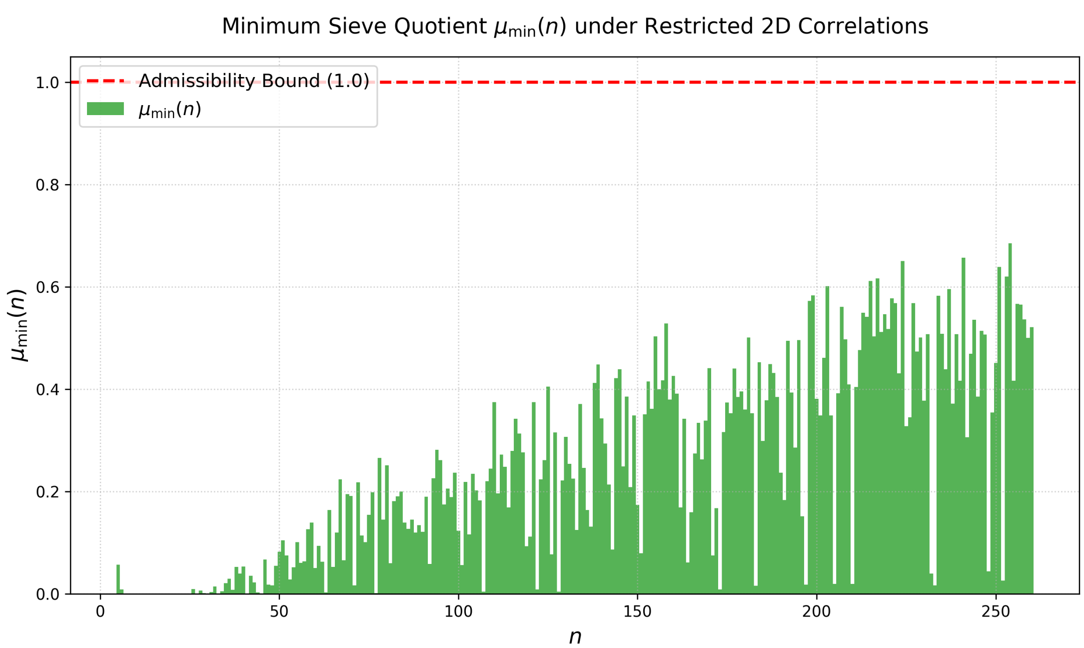

FIGURE 1. The minimum sieve quotient $\mu_{\min}(n)$ under restricted 2D correlations. The upward trend indicates that pairwise interference struggles to maintain the admissibility bound $\mu < 1$ for larger n.

To demonstrate the structural necessity of higher-order interactions to permanently breach the parity barrier, we performed a targeted experiment at $n = 199$. In the purely 2D model utilizing the complete prime basis for $n = 199$, the optimization yields a quotient of $\mu_{\min}(199) \approx 0.583$.

We subsequently relaxed the correlation depth to 3D, while restricting the basis to a highly active subset of 20 prominent primes (to keep the matrix dimension computationally feasible at $D = 1351$). The inclusion of 3D correlations sharply reduced the minimum quotient to $\mu_{\min}(199) \approx 0.081$. This sharp reduction corroborates the core analytic thesis of this paper: while 1D and 2D components struggle against the logarithmic divergence of the local penalty, higher-dimensional cross-correlations provide sufficiently large negative feedback to permanently satisfy the condition $\mu_{\min}(n) < 1$.

6.3. **Towards bounding the minimum ratio.** The exact resolution of this convolution computationally requires navigating the massive $2^{w(n)}$-dimensional polynomial space, making direct computation infeasible for large n. However, establishing the Sieve Sufficiency Guarantee unconditionally only requires demonstrating that $\mu_{\min}(n) < 1$. By the continuous variational principle, we are not required to locate the exact optimal minimizer λ_{opt}. Any analytically constructed test vector λ_{test} corresponding to a weight matrix that pushes the sieve weight quotient strictly below 1 is sufficient.

Advancing the resolution of the twin prime conjecture within this additive sieve framework therefore requires the following analytical steps:

(1) Expand the theoretical multidimensional convolution Ω_h and extract the lowest-order, highest-magnitude cross-correlation coefficients.
(2) Map these theoretical phase corrections onto the finite-dimensional polynomial basis $B_S(x) = \prod_{i \in S} \cos(6\pi x/p_i)$.
(3) Construct a continuous bounded weight vector λ_{test} over the target interval $\mathcal{A}_n$.
(4) Analytically bound the quadratic limits $Q_2(\lambda_{\text{test}})/Q_1(\lambda_{\text{test}})$, proving that the scaled cross-correlations possess the capacity to overcome the diverging spectral deficit.

While executing this final step demands careful estimation of the resulting combinatorial sums, the framework nevertheless provides a well-defined approach. Importantly, this continuous analytic framework maps a combinatorial obstacle into an explicit, seemingly resolvable optimization problem.

7. Conclusion

This paper has presented a rigorous reduction that equates the existence of twin primes to the bounded eigenvalue optimization of a specific residue structure. By tracking localized sieve penalties over symmetric bounded intervals and translating them into the frequency domain, we mapped discrete prime alignments to continuous character sum spectra.

This formulation allowed us to isolate the exact analytic manifestation of the parity barrier. We proved that one-dimensional exponential sum weights suffer from a structural deficit, forcing the sieve quotient $\mu \geq 1$ and formally necessitating the use of multidimensional polynomial bases. Combining topological compactness with harmonic analysis on geometric intervals, our reduction to $\mu_{\min}(n) < 1$ replaces the classical combinatorial parity obstruction with a concrete spectral bound. It refines the twin prime problem entirely into estimating the spectral density of heavily structured higher-order correlations, providing a concrete variational framework for addressing the parity barrier. All structural lemmas and equivalence bounds supporting this reduction have been machine-verified in Lean 4.

AI usage disclosure

The author acknowledges the use of generative AI (Gemini 3.1 Pro [7]) for LaTeX technical synthesis and structural formatting, as well as for structuring the Lean 4 formalization by designing precise prompt specifications for implementation by the Aristotle [1] automated theorem prover. The core mathematical logic, proofs, and final validation were performed by the author.

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